

GED® Math — Formula Cheat Sheet

Mathematical Reasoning · About 46 questions · 115 minutes · Passing score: 145/200 · certpathpublishing.store

NUMBER SENSE & PERCENTS

Fractions, Percents, Ratios

$$x\% = x/100 \quad \cdot \quad \text{"of"} = \times$$

- **% change** = (new – old) / old $\times$ 100
- **Part** = percent $\times$ whole; **Whole** = part / percent
- **Proportion**: $a/b = c/d \rightarrow$ cross-multiply ($ad = bc$)
- **Unit rate**: divide to get "per 1" (miles/hour, \$/item)
- Order of operations: **PEMDAS**

Exponents & Roots

$$a^m \cdot a^n = a^{m+n} \quad a^m / a^n = a^{m-n}$$

- $(a^m)^n = a^{mn}$ · $a^0 = 1$ · $a^{-n} = 1/a^n$ ($a \neq 0$)
- $\sqrt{25} = 5$ — $\sqrt{\quad}$ gives the principal (nonnegative) root
- But if $x^2 = 25$, then $x = \pm 5$ ($x^2 = k \rightarrow x = \pm\sqrt{k}$, $k \geq 0$)
- **Scientific notation**: $a \times 10^n$, with $1 \leq a < 10$
- Negative $\times$ negative = positive; neg $\times$ pos = negative

ALGEBRA & COORDINATE GEOMETRY

Lines & Slope

$$m = (y_2 - y_1) / (x_2 - x_1)$$

- **Slope-intercept**: $y = mx + b$ (b = y-intercept)
- **Point-slope**: $y - y_1 = m(x - x_1)$
- **Standard form (linear)**: $Ax + By = C$
- **Parallel** = same slope; **Perpendicular** = negative reciprocal (a horizontal line $\perp$ a vertical line)

Quadratics & Factoring

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

- **Standard form (quadratic)**: $y = ax^2 + bx + c$
- **Difference of squares**: $a^2 - b^2 = (a + b)(a - b)$
- **FOIL**: $(x + p)(x + q) = x^2 + (p+q)x + pq$
- **Distance between two points**: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- **Midpoint**: $((x_1 + x_2)/2, (y_1 + y_2)/2)$

GEOMETRY & MEASUREMENT

Area & Perimeter

- **Square**: $A = s^2$ · $P = 4s$
- **Rectangle**: $A = lw$ · $P = 2(l + w)$
- **Triangle**: $A = \frac{1}{2}bh$ · $P = s_1 + s_2 + s_3$ · angles = 180°
- **Parallelogram**: $A = bh$ · **Trapezoid**: $A = \frac{1}{2}(b_1 + b_2)h$
- **Circle**: $A = \pi r^2$ · $C = 2\pi r = \pi d$ ($\pi \approx 3.14$)
- **Pythagorean**: $a^2 + b^2 = c^2$

Volume

- **Any prism / box**: $V = Bh = lwh$ (B = base area)
- **Cylinder**: $V = \pi r^2 h$
- **Cone**: $V = \frac{1}{3}\pi r^2 h$
- **Pyramid**: $V = \frac{1}{3}Bh$ (B = base area)
- **Sphere**: $V = \frac{4}{3}\pi r^3$

SURFACE AREA, DATA & PROBABILITY

Surface Area

Total area of all outside surfaces — use the flat-face *and* curved-surface formulas (l = slant height):

- **Rectangular prism:** $SA = 2(lw + lh + wh)$
- **Cylinder:** $SA = 2\pi r^2 + 2\pi rh$
- **Cone:** $SA = \pi r^2 + \pi rl$
- **Sphere:** $SA = 4\pi r^2$
- **Pyramid:** $SA = B + \frac{1}{2}(\text{perimeter} \times l)$

Statistics & Probability

$$\text{Mean} = \text{sum} \div \text{count}$$

- **Median:** middle of sorted data (average the two middle if even count)
- **Mode:** most frequent · **Range:** $\text{max} - \text{min}$
- **Probability** (equally likely outcomes): $P = \text{favorable} \div \text{total}$, a value from 0 to 1
- **P(not A)** = $1 - P(A)$

RATES, MONEY & TEST DAY

Rates & Money

- **Distance:** $d = rt$ (distance = rate $\times$ time)
- **Total cost** = (number of units) $\times$ (price per unit)
- **Simple interest:** $I = prt$ (principal $\times$ rate $\times$ time)

Test-Day Facts

- About **46 questions** in **115 minutes** (~2.5 min each), with a short break between parts.
- **Part 1** (~5 questions): **no calculator**, comes first.
- **Part 2** (~41 questions): on-screen **TI-30XS MultiView** allowed.
- You need a **scaled score of 145/200** to pass. 145 is a scaled score — the number of correct answers needed varies by test form and is not a fixed percentage.
- A formula sheet is provided on-screen, but knowing the common formulas cold saves time.